

ABHINAV REJA

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SUMMARY

Software Engineering graduate student (MS, ASU) with 1.5+ years of experience in backend systems, agentic AI, and cloud infrastructure. Deployed AWS pipelines (IaaS, Serverless, Edge) with sub-second latency; fine-tuned Llama-3-8B and won HackASU 2025 for an agentic developer tool. Driven by system performance, LLM tooling, and scalable backend design.

EDUCATION

M.S. in Software Engineering Arizona State University, Arizona, USA	May 2026 GPA: 4.0/4.0
Bachelor's in Information Technology Oriental Institute of Science and Technology, Bhopal, India	May 2022 GPA: 3.36/4.0

EXPERIENCE

Backend Developer Intern, Vosyn Inc., Chicago, IL	June 2025 - August 2025
- Developed a history-aware content recommendation API powered by a fine-tuned Llama-3-8B (LoRA) classifier across 50+ genres, using batch inference and cached lookups to improve CTR by 15% (A/B tested). - Redesigned PostgreSQL schema for audio/video/text content tables using composite indexes and normalized relations, reducing p90 retrieval latency by ~30% across 3+ downstream services. - Engineered a Django REST batch-processing pipeline for bulk audio/video ingestion, reducing ingestion time by 11% and decoupling media processing from synchronous request cycles. - Implemented a Java/Cucumber integration testing framework integrated into the GitHub Actions CI/CD pipeline for API and workflow validation, reducing regression testing time per release cycle by 15%.	
Software Engineer, WINspect Technologies Pvt. Ltd., Mumbai, India	January 2023 - May 2024
- Built C++ REST APIs for a time-sync dashboard serving 500+ distributed clients, maintaining sub-200ms p95 latency. - Introduced a Redis caching layer with short TTL for synchronization data, reducing average API read latency by 50% for high-frequency polling requests. - Automated 40+ integration tests for synchronization features using Google Test (gtest), integrated into the Jenkins CI/CD pipeline, reducing manual testing effort by 30%. - Wrote Python automation scripts to remediate recurring production failures, reducing ticket resolution time by 18%.	

PROJECTS

CodeX HackASU 2025 Winner - Claude Developer Tools Track Devpost	November 2025
- Shipped an LLM-powered tool that auto-migrates backend projects across Flask, Django, and Spring Boot, winning the Claude Developer Tools track against 100+ submissions. - Designed a multi-stage agentic pipeline using Gemini and Claude APIs to analyze project structure, extract dependency graphs, and generate framework-idiomatic migrations in under 90 seconds per project. - Configured a Flask backend with persistent state and resumable sessions; implemented human-in-the-loop validation to flag ambiguous dependency conflicts before code generation.	
Face Recognition on AWS IaaS, Serverless, and Edge	September 2025 - November 2025
- Architected an IaaS pipeline with a custom autoscaler, S3/SQS queue, and PyTorch workers on a custom AMI, achieving 0.96s latency and 80% recognition accuracy across 100 concurrent requests. - Delivered a serverless variant using ECR, MTCNN, and SQS-triggered FaceNet inference, achieving 1.78s latency. - Deployed edge inference using AWS Greengrass and MQTT with Lambda-backed recognition, achieving 0.78s latency.	
WIAL Platform Opportunity Hack 2026 Demo · Code	March 2026
- Powered cross-lingual semantic search using OpenAI embeddings and pgvector, matching coaches across 20+ international chapters in 10+ languages via GPT-4o-mini query parsing. - Rebuilt fragmented WordPress sites into a unified Next.js app with service worker caching and AVIF/WebP assets, achieving sub-4s load times on slow 3G for 20+ global chapters. - Orchestrated multi-tenant chapter provisioning with Stripe payments and role-based notifications; GPT-4o-mini auto-generated website drafts for 20+ chapters in under 60 seconds each.	

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript
Frameworks: Flask, FastAPI, Django, Next.js, React, Spring Boot, Cucumber, Streamlit, Tailwind CSS
Cloud/DevOps: AWS (EC2, S3, SQS, Lambda, ECR, Greengrass), Docker, Jenkins, GitHub Actions, Vercel
Databases: PostgreSQL, MongoDB, Redis, pgvector
AI/ML: OpenAI API, Claude API, Gemini API, Llama-3 (LoRA fine-tuning), MTCNN, FaceNet, PyTorch